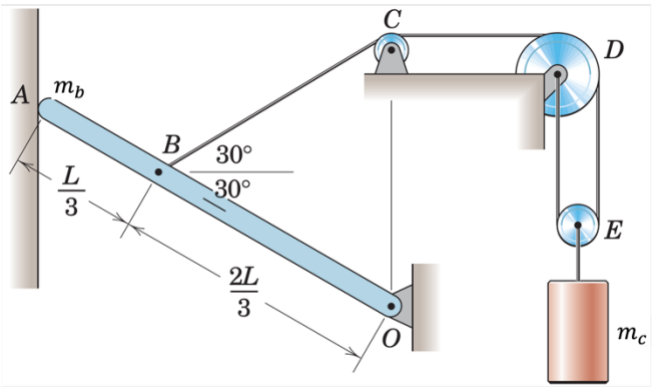
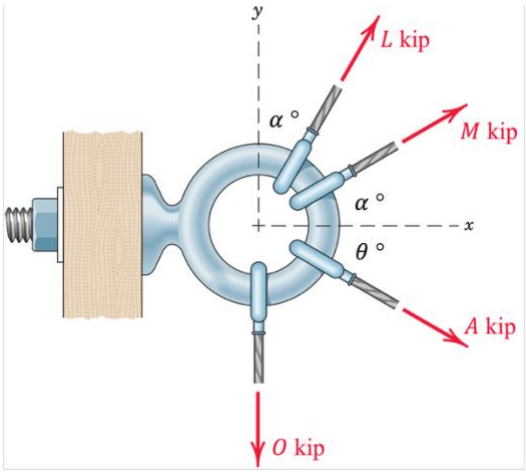
Quiz 03. Picking at the scabs of Exam 01
prepared by C. Cahig *for* ENS161 (AY 2026-2027, S1)

Instructions. Unless otherwise stated, all quantities are in SI units. Take the acceleration due to gravity to be 9.80665.

Part I. For identification questions, write answers in lowercase and, where multiple answers or values are required, separated by semicolon. Each item required is worth two (2) points.

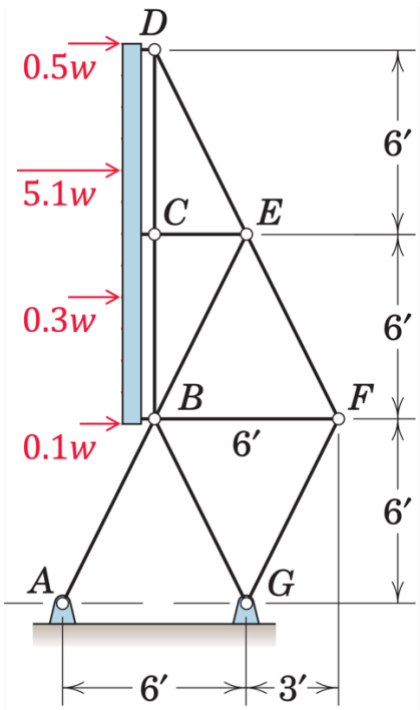
1. A static system of forces and/or moments acting on a rigid body is called _____ if the unknown support reactions can be found by using only the equilibrium equations.
2. An object is idealized as a/an ___ if its dimensions are deemed negligible.
3. *True or False.* If a rigid body is in static equilibrium, then the total moment due to the forces (including support reactions) about a point outside the body is zero.
4. Two cables sustain 12- and 14-lb tensions when loaded with a 1-tonne mass. What would be the cable tensions if the mass is halved?
5. An object at rest may be analyzed as a particle even if its dimensions are not negligible, provided that the forces acting on it are ___.
6. Two systems of forces and moments cause the same net force and net moment on an object. The systems are said to be ___.
7. *True or False.* Two forces having equal magnitudes but opposing directions form a couple.
8. With this theorem, the resultant moment of a system of forces about an axis is the algebraic sum of the moments of each of the forces about the same axis.
9. *True or False.* Concurrent forces are not always coplanar.
10. The principle of transmissibility of forces effectively treats a force as a/an ____ vector.
11. *True or False.* A statically determinate system is always stable.
12. *True or False.* The moment of a force $\boldsymbol{F}$ about a point A is perpendicular to both $\boldsymbol{F}$ and the position vector from A to any point in the line of action of $\boldsymbol{F}$.
13. A force vector is characterized by its ___, ___, and ___.
14. In the following diagram, m_b and m_c are the respective masses of the bar and of the cylinder. For what m_c-to-m_b ratio/s will the bar becomes a three-force member in static equilibrium.





Scenario 1. In Alan Parsons' project, an eyebolt in the skyroof system anchors four cables. Suppose that the forces at the perspective of the eyebolt are approximately coplanar, with $\alpha = 30$, $L = 2$, $A = 1.2$, and $O = 1.8$. The four forces must have a purely horizontal resultant. The eyebolt-to-wood joint can sustain no more than 3 kips. Solve for:

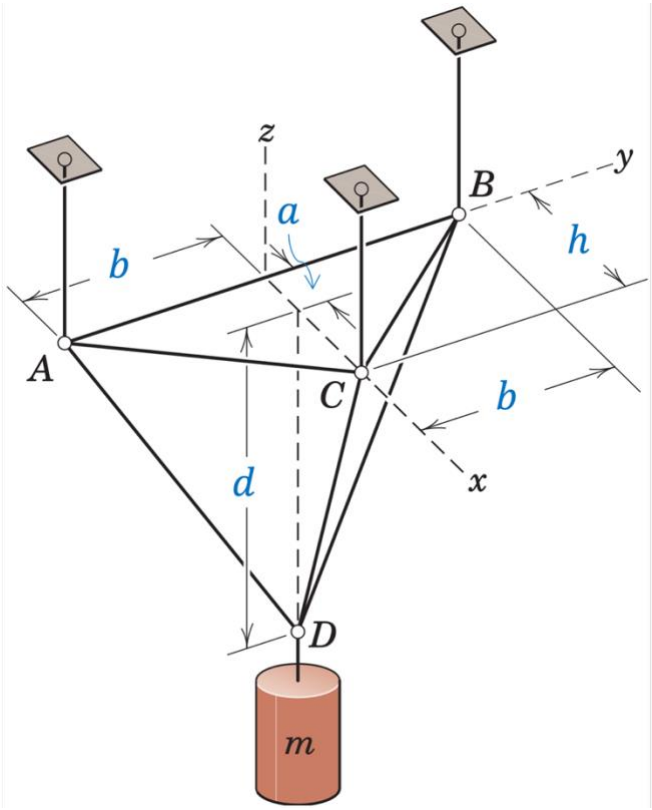
a. [10 pts] the limiting value (in kips) of M, and
b. [10 pts] the limiting value (in degrees) of θ.



Scenario 2. To befriend the will of the wind is to have one less truss issue. Here, an erratic wind load case is modelled as $0.5w$-, $5.1w$-, $0.3w$-, and $0.1w$-lb forces evenly spaced along the height of the signboard. The truss tower is supported by cable AB and a pin at G. Each of the nine segments of the tower weighs 2 lb/ft, while the signboard weighs 3 lb/ft. Assume that the combined center of gravity of the tower and the signboard is at B. If it is unsafe for cable AB to carry tensions more than 67 lb, solve for:

a. [10 pts] the maximum allowable value of w, and
b. [20 pts] the reaction (as a Cartesian vector) at G corresponding to the maximum allowable value of w.

Quiz 03. Picking at the scabs of Exam 01
prepared by C. Cahig *for* ENS161 (AY 2026-2027, S1)



Scenario 3. An m-kg model load is hangin' by a thread but is supported by cables AD, CD, and BD. Set a, b, h, and d to 1, 4, 3, and 6 m, respectively. For every kilogram of the model load, determine:

a. [4 pts] the moment (as a Cartesian vector) about C due to the load,
b. [5 pts] the moment (as a Cartesian vector) about the line through A and B due to the load, and
c. [3 pts] the tension T_{AD} developed in cable AD,
d. [3 pts] the tension T_{CD} developed in cable CD, and
e. [3 pts] the tension T_{BD} developed in cable BD.

If cables AD, CD, and BD have respective strengths of 76, 67, and 76 N, solve for:

f. [6 pts] the maximum allowable model load mass,
g. [2 pts] the corresponding tension in cable AD,
h. [2 pts] the corresponding tension in cable CD, and
i. [2 pts] the corresponding tension in cable BD.

Marking. The total percentage rating a student gets for this Activity is computed as $0.59(0.9^m r) + 0.41p$, where m is the number of unmet requirements and violated instructions, r is the raw score (in percent), and p is the peer rating (in percent) by his/her groupmates.

Submission. Answers shall be communicated through the accompanying Google Form. Peer ratings shall be communicated via the submission platform in the Google Classroom.